

Quiz 1 - PHY 206, Physics Through Computational Thinking

(February 11, 2025)

Total Marks - 20

(10:05 - 10:50 AM)

Suppose your enrollment number is 23001

- Create your Mathematica file quiz-23001.nb and submit only this file
- At the time of submission **Zip the file** and upload it at <http://172.28.153.64:5000/>
- The uploading window will be available only for 5 minutes at the end of exam
- Please indicate the question number clearly in the answer sheet.
- Use of Mathematica documentation is allowed.
- Use of calculators, internet is not allowed.

Problem 1: Find the 100th digit of $\frac{\pi^2}{6}$ by writing appropriate Mathematica function. (1)

Problem 2: How many time does the curves $y = x^5$ and $y = e^{x/4}$ intersect each other. Find out the numerical values of all these intersection points and plot separate curves to show each intersection of the curves. (3)

Problem 3: Plot the function $-\frac{10}{x^5} + \frac{1}{x^6}$ and find whether it has a minima or not. In case you find the minima, perform a Taylor expansion of the function around the minima up to the quadratic term. Plot the original function and the Taylor approximation on the same graph. (3)

Problem 4: Plot the following functions $\frac{1}{x^2 - a^2}$ and analyse it for its zeros, divergences, extrema, and asymptotes. Write your comments about each of the points in your Mathematica Notebook. (3)

Problem 5: Plot the vector field $\dot{x} = r - x - x^3 + 0.5x^4$ by looking at the plot and discuss the criteria for existence of the equilibrium points. Also discuss the stability of the equilibrium points. (3)

Problem 6: A particle moves in a trajectory given by $r = A\theta$, where A is a constant. θ increases with time according to $\theta = \alpha t^2/2$, where α is a constant. Sketch the trajectory of the particle using non-dimensionalization concept discussed in the class. (4)

Problem 7: Consider the three point charges are kept at:

- $(-10, 0)$
- $(0, 0)$
- $(10, 0)$

Generate a **vector field plot** of the electric field generated by these in the xy -plane. Show the variation of electric field by varying the magnitude and sign of the charges. (3)